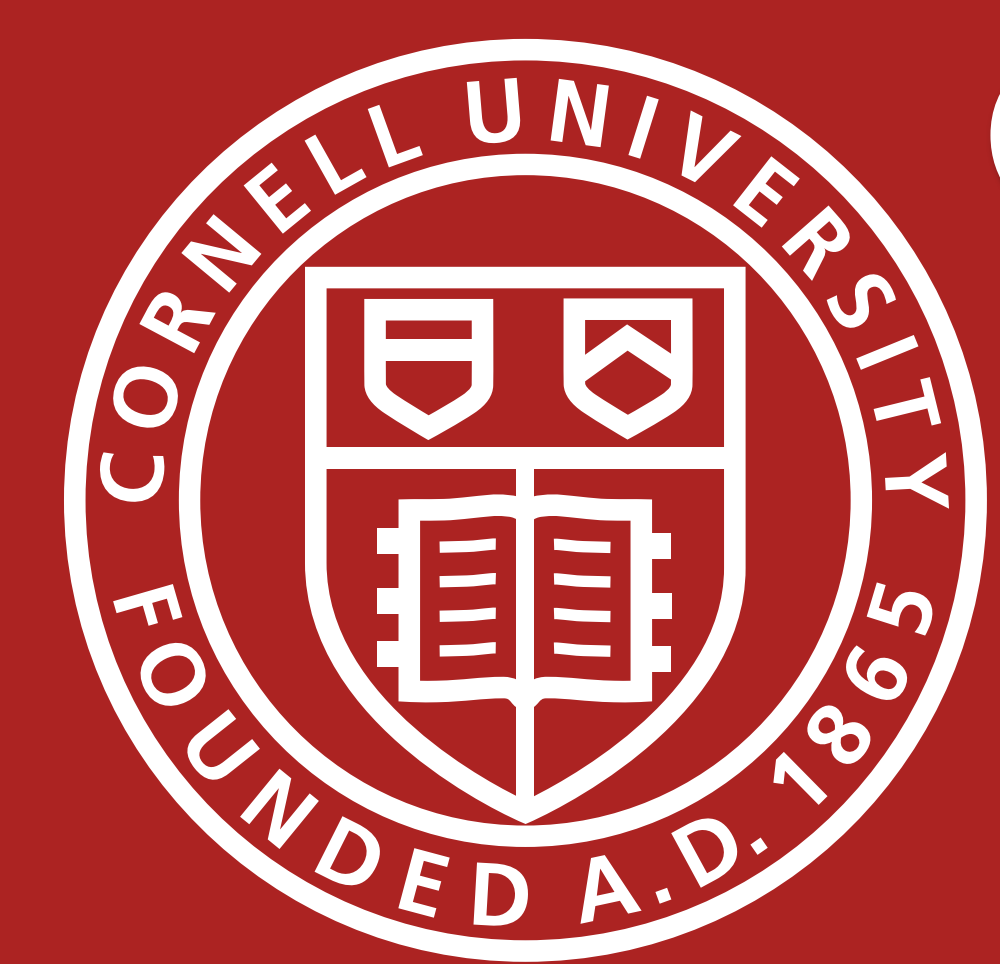




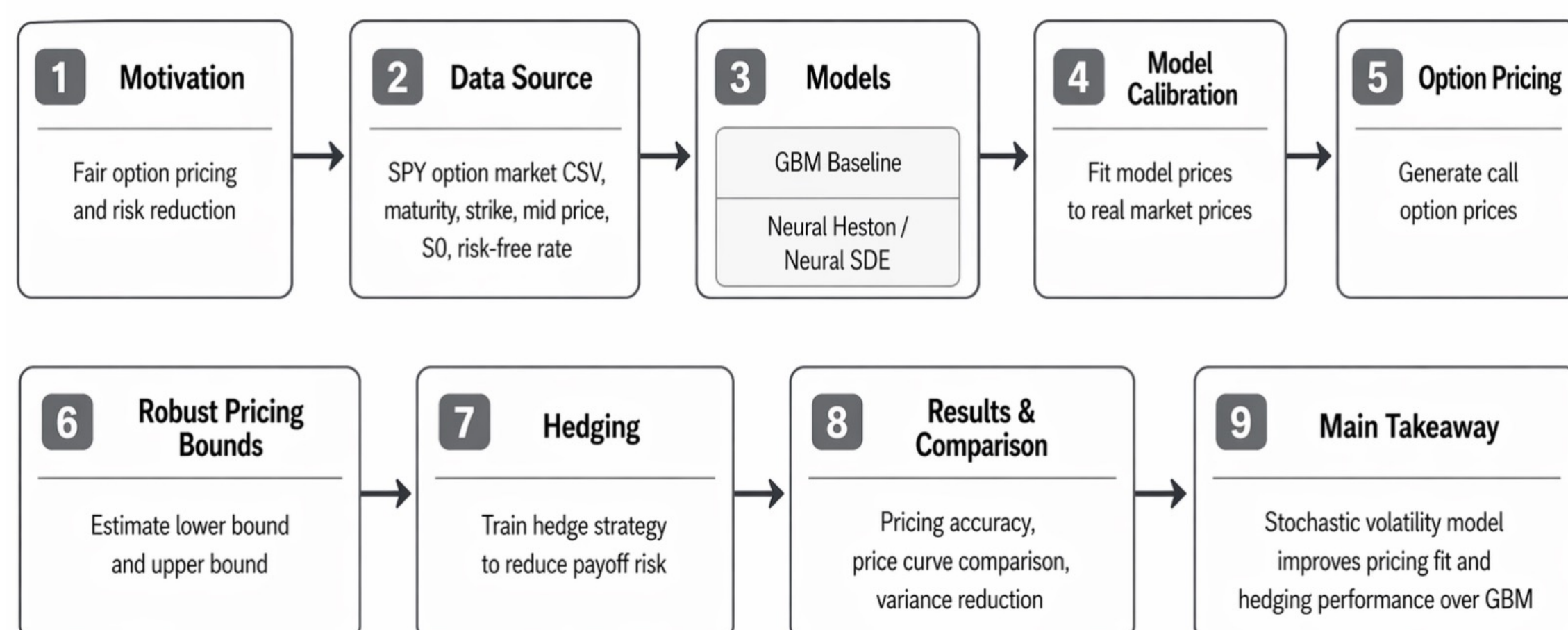
Option Pricing and Hedging with Neural Stochastic Differential Equation Model

Comparing Geometric Brownian Motion Model vs. NSDE Model on Real Option Data

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Motivation & Workflow

An option is a financial contract whose value depends on the future price of an underlying asset. In this project, we study European call options, which give the holder the right to buy the asset at a fixed price on the maturity date. Option pricing estimates the fair value of the contract, while hedging reduces the risk after taking the position. We use real SPY call option market data to compare a simple GBM baseline with a more flexible Neural Heston / Neural SDE model in pricing, robust pricing, and hedging.



Models

GBM Baseline

The GBM baseline assumes constant volatility:

$$dS_t = rS_t dt + \sigma S_t dW_t$$

S_t : asset price at time t

r : risk-free interest rate

σ : constant volatility

W_t : Brownian motion, representing random market shocks

dt : small time increment

Neural Heston / Neural SDE

The main model uses stochastic volatility:

$$dS_t = rS_t dt + \sqrt{V_t} S_t dW_t$$

$$dV_t = \kappa(\mu - V_t) dt + \eta \sqrt{V_t} dB_t$$

$$\text{corr}(dW_t, dB_t) = \rho$$

κ : mean-reversion speed

μ : long-run variance level

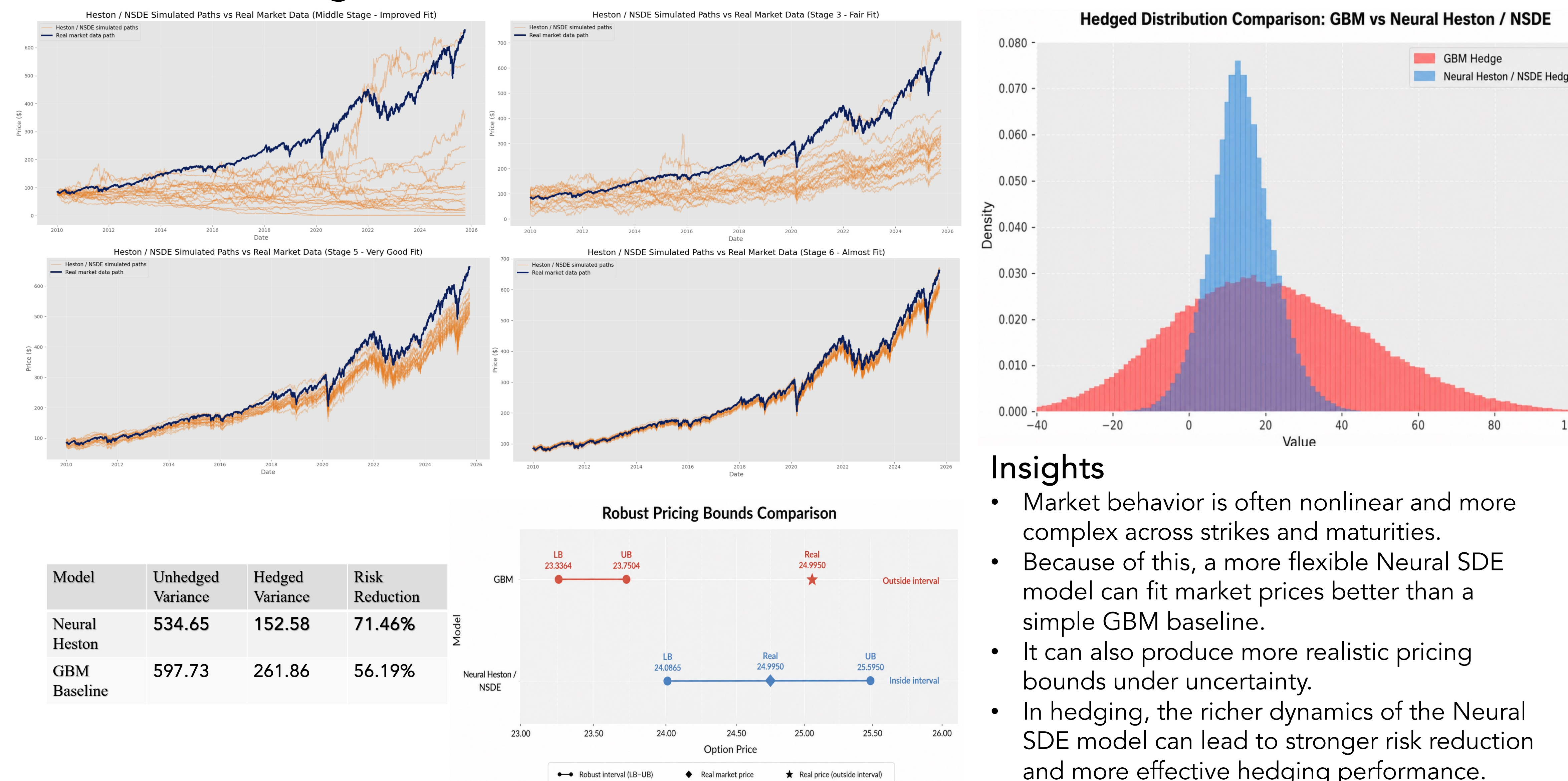
η : volatility of volatility

ρ : correlation between price and variance shocks

V_0 : initial variance

$$\theta_{\text{NSDE}} = (\kappa, \mu, \eta, \rho, V_0)$$

Results & Insights



Insights

- Market behavior is often nonlinear and more complex across strikes and maturities.
- Because of this, a more flexible Neural SDE model can fit market prices better than a simple GBM baseline.
- It can also produce more realistic pricing bounds under uncertainty.
- In hedging, the richer dynamics of the Neural SDE model can lead to stronger risk reduction and more effective hedging performance.

Hedging

Pricing alone is not enough because the final payoff remains uncertain. We hedge to reduce this payoff risk over time.

H: discounted option payoff.

G: cumulative hedge gain

R: final hedged residual risk

$$H = e^{-rT}(S_T - K)^+ \quad \bar{S}_t = e^{-rt}S_t$$

$$G = \sum_{k=0}^{n-1} \Delta t_k (\bar{S}_{t_{k+1}} - \bar{S}_{t_k}), \quad R = H - G$$

A good hedge should reduce the residual variance.

Risk Reduction measures how much the hedge lowers the payoff variance. A larger positive value means a more effective hedge.

$$\text{Var}(R) < \text{Var}(H)$$

$$\text{Risk Reduction} = \left(1 - \frac{\text{Var}(R)}{\text{Var}(H)}\right) \times 100\%$$

Advanced hedging objective

We learn hedge parameters by minimizing the conditional variance of the hedging error:

$$\bar{\xi}^* \in \arg \min_{\bar{\xi}} \text{Var} \left[\Phi((X_t)_{t \in [0, T]}) - \int_0^T \bar{h}(r, (X_{r \wedge t})_{t \in [0, T]}, \bar{\xi}) d\bar{S}_r^\theta \middle| \mathcal{F}_0 \right]$$

The term $\Phi((X_t)_{t \in [0, T]})$ represents the target payoff as a function of the state path, while $\int_0^T \bar{h}(\cdot, \bar{\xi}) d\bar{S}_r^\theta$ represents the cumulative gain from the learned hedge strategy. This means the hedge is trained to make the final residual risk as small as possible.

Variable Meanings

- $\bar{\xi}$: trainable hedge parameters
- $\bar{\xi}^*$: optimal hedge parameters
- $\Phi((X_t)_{t \in [0, T]})$: option payoff / target claim
- X_t : state process over time
- $\bar{h}(\cdot, \bar{\xi})$: hedge policy produced by the neural network
- $\bar{S}_t^\theta$: discounted asset price under model parameters θ
- $\mathcal{F}_0$: information available at the initial time
- $\text{Var}[\cdot | \mathcal{F}_0]$: conditional variance of the hedging error

Calibration

We calibrate both models to the same real SPY call option dataset, so the comparison is fair:

- GBM baseline
- Neural Heston / Neural SDE

The calibration inputs are the observed option contracts $(T_i, K_i, P_i^{\text{market}})$, where T_i is maturity, K_i strike, and P_i^{market} is the market mid price.

Calibration inputs are the observed option contracts using mean squared error, calibration:

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N (P_{\theta}(T_i, K_i) - P_i^{\text{market}})^2$$

The pricing formula is: $P(T, K) = e^{-rT} \mathbb{E}[(S_T - K)^+]$

After calibration, both models are used to generate option prices. Since the Neural Heston / NSDE model does not have a simple closed-form solution here, prices are estimated by Monte Carlo simulation:

$$P(T, K) \approx e^{-rT} \frac{1}{M} \sum_{m=1}^M \max(S_T^{(m)} - K, 0)$$

Robust Pricing

After calibration, the model is used to price a new option with maturity T^* and strike K^*

$$P_{\theta}(T^*, K^*) = e^{-rT^*} \mathbb{E}[(S_{T^*} - K^*)^+]$$

Here, θ is the calibrated parameter vector, T^* is the target maturity, and K^* is the target strike.

Because multiple parameter sets may fit market prices similarly well, a single calibrated price may not fully capture model uncertainty. We therefore compute a lower bound and upper bound:

$$\text{Upper \& Lower bound: } \min_{\theta} [\pm P_{\theta}(T^*, K^*) + \lambda \mathcal{L}_{\text{calib}}(\theta)]$$

This produces a pricing interval for the new option under calibration uncertainty.

Variable Meanings

- T^* : maturity of the new target option
- K^* : strike of the new target option
- θ : model parameter vector
- $P_{\theta}(T^*, K^*)$: model-implied price of the new option
- $\mathcal{L}_{\text{calib}}(\theta)$: calibration loss
- λ : penalty weight controlling the tradeoff between fit and price exploration

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